Assignment Problem

Kolagani Paramahamsa

May 8, 2026

PRELIMINARY DRAFT

1 Cost Minimization

A bakery+cafe company owns bakeries $\mathcal{X}$ and cafes $\mathcal{Y}$. Let $|\mathcal{X}| = |\mathcal{Y}| = n$. The bakeries have a supply of one bread each, and the cafes demand one bread each. The cost of transporting bread from bakery x_i to cafe y_j is $c(x_i, y_j)$. The bakery+cafe company's objective is to find a bijective function (an assignment) $T : \mathcal{X} \to \mathcal{Y}$ that minimizes total cost,

$$\min_{T \text{ bijective}} \sum_{i=1}^{n} c(x_i, T(x_i)).$$

Existence of an optimal T is not an issue in this discrete problem, as we can check all the finitely many possible Ts.

1.1 Cyclical monotonicity

Definition 1. An assignment T is c-cyclically monotone if for any $k \leq n$, and any family $(x_1, y_1), \ldots, (x_k, y_k) \in \{(x_i, y_i) : y_i = T(x_i)\}$ satisfies the inequality

$$\sum_{i=1}^{k} c(x_i, y_i) \leq \sum_{i=1}^{k} c(x_{i+1}, y_i),$$

with the convention that $x_{i+1} = x_1$.

Lemma 1. *An assignment T is optimal if and only if it is c-cyclically monotone.*

Proof. If T is not c-cyclically monotone, then there exist $(x_1, y_1), \ldots, (x_k, y_k)$ in the graph of T such that

$$\sum_{i=1}^{k} c(x_i, y_i) > \sum_{i=1}^{k} c(x_{i+1}, y_i),$$

where $x_{k+1} = x_1$.

Define a new assignment S by

$$S(x_i) = y_{i-1}, \qquad i = 1, \ldots, k,$$

(with $y_0 = y_k$), and $S(x) = T(x)$ for all other x. Then S is still bijective, and

$$\sum_{i=1}^{n} c(x_i, S(x_i)) < \sum_{i=1}^{n} c(x_i, T(x_i)).$$

Hence T is not optimal.

Conversely, suppose T is c-cyclically monotone. Let S be any other bijection. Consider the permutation

$$\pi := T^{-1} \circ S$$

of $\mathcal{X}$. As $|\mathcal{X}|$ is finite, every permutation decomposes into disjoint cycles. On each cycle

$$x_1 \mapsto x_2 \mapsto \cdots \mapsto x_k \mapsto x_1,$$

we have

$$S(x_i) = T(x_{i+1}).$$

Applying c-cyclical monotonicity gives

$$\sum_{i=1}^{k} c(x_i, T(x_i)) \leq \sum_{i=1}^{k} c(x_i, T(x_{i+1})) = \sum_{i=1}^{k} c(x_i, S(x_i)).$$

Summing over all disjoint cycles yields

$$\sum_{i=1}^{n} c(x_i, T(x_i)) \leq \sum_{i=1}^{n} c(x_i, S(x_i)).$$

Since S was arbitrary, T is optimal. $\square$

Remark 1. The definition of c-cyclical monotonicity gives a structure to rearrangements. First choosing a cycle length, then choosing any subset of pairs with that length. Since this structure accounts for all the rearrangements, we can just focus on it for our analysis.

2 The Dual Problem

The Story. Consider a logistics company that can magically (read zero cost) transport bread between bakeries and cafes. It can offer to buy bread at bakeries and sell at cafes - at some prices. The is the only contract space available to the logistics company. Also assume that this contract is non-binding in the following sense: the bakery+cafe company has power to unilaterally deviate by refusing to trade at some bakery and cafe and transport it themselves[1] Taking this into account the logistics company ensures prices are such that this scenario won't occur.

Let $\psi : \mathcal{X} \to \mathbb{R}$ and $\phi : \mathcal{Y} \to \mathbb{R}$ price functions at bakeries and cafes, respectively. The logistics company's revenue-maximization problem (the dual) is then

$$\begin{aligned} &\max_{\psi,\phi} \sum_{j=1}^{n} \phi(y_j) - \sum_{i=1}^{n} \psi(x_i), \\ &\text{s.t. } \phi(y_j) - \psi(x_i) \leq c(x_i, y_j) \qquad \forall\, i, j \in \{1, \ldots, n\}. \end{aligned}$$

Lemma 2. [Weak Duality]

$$\max_{\phi-\psi\leq c} \sum_{y_j=1}^{n} \phi(y_j) - \sum_{x_i=1}^{n} \psi(x_i) \leq_T \min_{\textit{bijective}} \sum_{i=1}^{n} c(x_i, T(x_i)).$$

[1]If logistics company can make the bakery+cafe company to sign a binding contract it is enough for logistics company to guarantee total loss for cafe+bakery company is less than the optimal transport cost.

Proof. Fix any feasible T (bijective). For any feasible ψ, ϕ in the dual, we have $\phi(y_i) - \psi(x_i) \leq c(x_i, y_j)$ for all $i, j \in \{1, \ldots, n\}$. In particular $\phi(T(x_i)) - \psi(x_i) \leq c(x_i, T(x_i))$. Summing over i, we see that LHS is equivalent to the dual objective since T is bijective. The RHS sums to the primal objective. Therefore, every feasible dual objective is less than every feasible primal objective. The result then follows. □

Despite simple proof of weak duality, it is not obvious that there exist prices such that the two objectives meet. The logistics company sets competitive prices between *all* pairs of bakeries and cafes, not just the ones between which bakery+cafe company would have transported in its optimal plan. Despite the restrictive constraints, we show that the optimal objectives match.

Theorem 1. [Strong Duality]

$$\max_{\phi - \psi \leq c} \sum_{y_j=1}^{n} \phi(y_j) - \sum_{x_i=1}^{n} \psi(x_i) = \min_{T \text{ bijective}} \sum_{i=1}^{n} c(x_i, T(x_i)).$$

Proof. Due to Lemma 2, the strong duality can only hold at primal's optimal, fix such a T. We denote the graph of T by Γ. Without loss of generality, assume that $(x_i, y_i) \in \Gamma$ for all $i \in \{1, \ldots, n\}$. To have any hope for strong duality we assume prices satisfy the following equations,

$$\phi(y_i) - \psi(x_i) = c(x_i, y_i) \qquad \forall i \in \{1, \ldots, n\}. \tag{1}$$

Now that we matched the objectives we need to make sure the constraints in the dual are satisfied,

$$\phi(y_j) - \psi(x_i) \leq c(x_i, y_j) \qquad \forall i, j \in \{1, \ldots, n\}. \tag{2}$$

Substituting equation (1) with $i = 1$ into inequality (2) with $i = 2, j = 1$ we get

$$\psi(x_2) \geq \psi(x_1) + [c(x_1, y_1) - c(x_2, y_1)].$$

Fix some $(x_1, y_1) \in \Gamma$. Pick any $m \geq 2$ and any $(x_2, y_2), \ldots, (x_m, y_m) \in \Gamma$. Then doing the above exercise recursively we derive for any $x \in \mathcal{X}$.

$$\psi(x) \geq \psi(x_1) + [c(x_1, y_1) - c(x_2, y_1)] + \ldots + [c(x_m, y_m) - c(x, y_m)]$$

We assume without loss of generality that $\psi(x_1) = 0$ since the dual problem remains unchanged if we subtract $\psi(x_1)$ from all prices ψ and ϕ. Motivated by above necessary inequality we define,

$$\psi(x) := \sup_{m} \sup_{(x_2,y_2),\ldots,(x_m,y_m)\in\Gamma} [c(x_1, y_1) - c(x_2, y_1)] + \ldots + [c(x_m, y_m) - c(x, y_m)]$$

Adding an subtracting $c(x_1, y_m)$ we get

$$\begin{aligned}\psi(x) = \sup_{m} \sup_{(x_2,y_2),\ldots,(x_m,y_m)\in\Gamma} &\{[c(x_1, y_1) - c(x_2, y_1)] + \ldots + [c(x_m, y_m) - c(x_1, y_m)]\} + [c(x_1, y_m) - c(x, y_m)] \\ \leq [c(x_1, y_m) - c(x, y_m)]&,\end{aligned}$$

where the inequality is true due c-cycle monotonicity. Hence, ψ is finite and well-defined.

We then define ϕ using equation (1). To verify inequality (2) pick any i, j then

$$\phi(y_j) - \psi(x_i) = \psi(x_j) - \psi(x_i) + c(x_j, y_j) \leq c(x_i, y_j).$$

The inequality is true because from definition of ψ we have

$$\begin{aligned}\psi(x_i) \geq \sup_{m-1} & \sup_{(x_2,y_2),\ldots,(x_{m-1},y_{m-1})\in\Gamma} \{[c(x_1,y_1) - c(x_2,y_1)] + \ldots + [c(x_{m-1},y_{m-1}) - c(x_j,y_{m-1})]\} \\ & + [c(x_j,y_j) - c(x_i,y_j)] \\ & = \psi(x_j) + [c(x_j,y_j) - c(x_i,y_j)],\end{aligned}$$

where we considered sup among all chains that end in x_i and pass through x_j in the last but one step. □

Remark 2. Although we have a finite problem, we used unbounded m to define ψ because we want to consistently define it such that *all* the inequality constraints are satisfied simultaneously. In the last step of the proof we exploit recursive nature of the construction. This is also precisely where we used c-cycle monotonicity property to show the definition is well-defined.